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Question	Answer	Marks
1	**Defining the problem**	
	R is the independent variable and *E* is the dependent variable **or** vary *R* and measure *E*	**1**
	keep *V* constant	**1**
	Methods of data collection	
	labelled diagram of workable experiment including: • coil P placed close to coil Q • separate workable circuit for coil Q • (a.c.) voltmeter or oscilloscope connected across coil Q (Do not accept a power supply connected to coil Q.)	**1**
	a.c. power supply/signal generator connected to resistor and coil P in series	**1**
	workable circuit with power supply and (a.c.) voltmeter/oscilloscope in parallel with resistor and coil P or across terminals of power supply/signal generator	**1**
	method to determine *R*, e.g. measure current in *R* and p.d. across *R* and use $R = V_R/I$ **or** measure *R* using an ohmmeter	**1**

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Question	Answer	Marks
1	**Method of Analysis**	
	plot a graph of $\frac{1}{E}$ against R or equivalent (e.g. R against $\frac{1}{E}$) Do not accept logarithms.	**1**
	$M = \frac{1}{2\pi fV \times \text{gradient}}$ (for R against $\frac{1}{E}$: $M = \frac{\text{gradient}}{2\pi fV}$)	**1**
	$k = 2\pi fVM \times y$-intercept **or** $k = \frac{y\text{-intercept}}{\text{gradient}}$ (for R against $\frac{1}{E}$: $k = -y$-intercept)	**1**